4 PREDICTOR

4.1 OVERVIEW

This section specifies the calculation of the *predicted sample values* $\hat{s}_{z,y,x}$ and *mapped quantizer indices* $\delta_{z,y,x}$ from the input image samples $s_{z,y,x}$. Quantities defined in this section are summarized in table E-2 of annex E.

This Recommended Standard makes use of the same adaptively weighted predictor as reference [D2], but to accommodate near-lossless compression, prediction calculations are performed using *sample representatives* $s''_{z,y,x}$, defined in 4.9, in place of the original sample values $s_{z,y,x}$. This is necessary so that the decompressor can duplicate the prediction calculation.

Prediction can be performed causally in a single pass through the image. Prediction at sample $s_{z,y,x}$, that is, the calculation of $\hat{s}_{z,y,x}$ and $\delta_{z,y,x}$, generally depends on the values of sample representatives for nearby samples in the current spectral band and P preceding (i.e., lower-indexed) spectral bands, where P is a user-specified parameter (see 4.2). Figure 4-1 illustrates the typical neighborhood used for prediction; this neighborhood is suitably truncated when $y = 0$, $x = 0$, $x = N_X - 1$, or $z < P$.

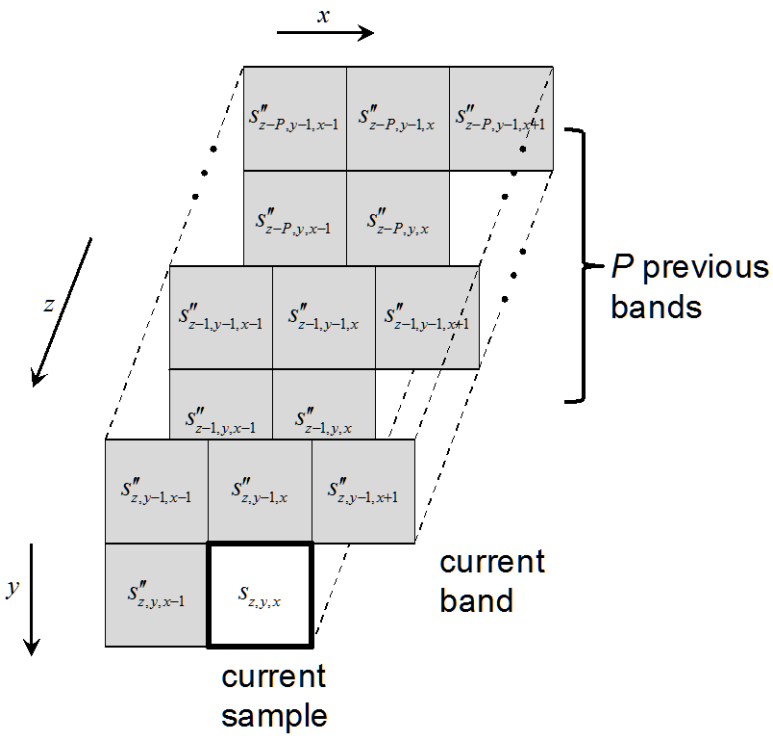

Figure 4-1: Typical Prediction Neighborhood

$$\omega_z^{(i)}(t+1) = \text{clip}\left(\omega_z^{(i)}(t) + \left\lfloor \frac{1}{2}\left(\text{sgn}^+\left[e_z(t)\right]\cdot 2^{-\left(\rho(t)+\varsigma_z^{(i)}\right)}\cdot d_{z-i}(t)+1\right)\right\rfloor, \left\{\omega_{\min}, \omega_{\max}\right\}\right), \tag{51}$$

and, when full prediction mode is used, for the directional components,

$$\omega_z^{\text{N}}(t+1) = \text{clip}\left(\omega_z^{\text{N}}(t) + \left\lfloor \frac{1}{2}\left(\text{sgn}^+\left[e_z(t)\right]\cdot 2^{-\left(\rho(t)+\varsigma_z^{*}\right)}\cdot d_z^{\text{N}}(t)+1\right)\right\rfloor, \left\{\omega_{\min}, \omega_{\max}\right\}\right), \tag{52}$$

$$\omega_z^{\text{W}}(t+1) = \text{clip}\left(\omega_z^{\text{W}}(t) + \left\lfloor \frac{1}{2}\left(\text{sgn}^+\left[e_z(t)\right]\cdot 2^{-\left(\rho(t)+\varsigma_z^{*}\right)}\cdot d_z^{\text{W}}(t)+1\right)\right\rfloor, \left\{\omega_{\min}, \omega_{\max}\right\}\right), \tag{53}$$

$$\omega_z^{\text{NW}}(t+1) = \text{clip}\left(\omega_z^{\text{NW}}(t) + \left\lfloor \frac{1}{2}\left(\text{sgn}^+\left[e_z(t)\right]\cdot 2^{-\left(\rho(t)+\varsigma_z^{*}\right)}\cdot d_z^{\text{NW}}(t)+1\right)\right\rfloor, \left\{\omega_{\min}, \omega_{\max}\right\}\right). \tag{54}$$

4.10.4 The inter-band weight exponent offsets $\varsigma_z^{(i)}$, for $z = 0,\ldots, N_Z - 1$ and $i = 1,\ldots,P_z^*$, and intra-band weight exponent offsets ς_z^* shall be user-specified integers in the range $-6 \le \varsigma_z^{(i)} \le 5$ and $-6 \le \varsigma_z^* \le 5$, respectively.

NOTE – The quantity $\left\lfloor \frac{1}{2}\left(\text{sgn}^+\left[e_z(t)\right]\cdot 2^{-(\rho(t)+\zeta)}\cdot d+1\right)\right\rfloor$ is equivalent to $\left\lfloor \frac{1}{2}\left(\left\lfloor \text{sgn}^+\left[e_z(t)\right]\cdot 2^{-(\rho(t)+\zeta)}\cdot d\right\rfloor+1\right)\right\rfloor$ but is not in general equivalent to $\left\lfloor \frac{1}{2}\left(\text{sgn}^+\left[e_z(t)\right]\cdot\left\lfloor 2^{-(\rho(t)+\zeta)}\cdot d\right\rfloor+1\right)\right\rfloor$.

4.11 MAPPED QUANTIZER INDEX

The signed quantizer index $q_z(t)$ is converted to an unsigned *mapped quantizer index* $\delta_z(t)$ defined as

$$\delta_z(t) = \begin{cases} \left|q_z(t)\right| + \theta_z(t), & \left|q_z(t)\right| > \theta_z(t) \\ 2\left|q_z(t)\right|, & 0 \le (-1)^{\tilde{s}_z(t)} q_z(t) \le \theta_z(t) \\ 2\left|q_z(t)\right| - 1, & \text{otherwise} \end{cases}, \tag{55}$$

where

$$\theta_z(t) = \begin{cases} \min\left\{\hat{s}_z(0) - s_{\min}, s_{\max} - \hat{s}_z(0)\right\} & t = 0 \\ \min\left\{\left\lfloor \frac{\hat{s}_z(t) - s_{\min} + m_z(t)}{2m_z(t)+1}\right\rfloor, \left\lfloor \frac{s_{\max} - \hat{s}_z(t) + m_z(t)}{2m_z(t)+1}\right\rfloor\right\}, & t > 0 \end{cases}. \tag{56}$$

NOTE – Each mapped quantizer index $\delta_z(t)$ can be represented as a D-bit unsigned integer.